

Supervision-Matched Selection of Paired RAG Repairs: An Empirical Study of Accuracy and Damage

Anonymous authors

Abstract

Retrieval repair can recover an incorrect answer but can also replace a correct answer. We study post-generation selection between original and repaired answers under shared candidates, matched supervision, and a fixed action allocation. The evaluation uses one Qwen2.5-3B-Instruct reader, 6,000 questions from three multi-hop QA datasets, three retrieval conditions, and nine policies. This design makes useful recoveries and harmful replacements visible under one intervention budget. A calibrated logistic policy combining a frozen internal HGB score with a paired Generate but Verify adaptation improves exact match by 0.2333 percentage points and reduces full-population Damage by 0.0722 points relative to the supervision-matched GbV-only policy; multiplicity-adjusted question-cluster bootstrap ranges are [0.0722, 0.4333] and [-0.1444, -0.0278]. The same joint rule does not pass against the HGB-only policy, and the 25-feature ROA-FULL policy does not advance over the two-signal policy. The result supports a bounded within-reader empirical attribution rather than a new selector architecture, reader-general transfer, or deployment-cost claim.

Keywords: retrieval-augmented generation; answer verification; selective prediction; reliability; empirical evaluation

1 Introduction

Retrieval-augmented generation (RAG) gives a language model access to external evidence, but retrieval is not uniformly helpful [12, 13, 17]. A second retrieval pass can supply missing evidence and recover an incorrect response. It can also distract the same reader and turn a correct answer into an incorrect one [1]. A repair mechanism therefore creates two separate questions: whether a useful alternative can be generated and whether that alternative should replace the original response.

This paper studies the second question after candidate acquisition. For each question and retrieval condition, a fixed pipeline produces an original answer a_0 and a repaired answer a_1 . A policy can keep a_0 or switch to a_1 . The empirical target is deliberately narrow: under shared candidates, matched supervision, and the same number of actions, does combining a frozen internal score with an adapted external verifier improve correctness while reducing damage relative to either signal alone?

The distinction matters because several broader ideas are established. Adaptive retrieval routes questions before or during generation [2, 9, 20]; verification can trigger retrieval and editing [22]; Answering with Faithfulness explicitly couples generation and faithfulness prediction [6]; and recent work predicts retrieval benefit or arbitrates between already generated answers [4, 11, 19]. Paired evidence interventions are also used to audit reader-conditional response [5]. Our two-input logistic policy is consequently an empirical comparison object, not a novel architecture.

We make three contributions. First, we evaluate five fitted policies under identical fit/calibration partitions, preprocessing, candidate pairs, eligibility, and global action count, while retaining four

additional contextual policies. Second, we report Recovery and Damage over the complete 18,000-trace population alongside exact match (EM) and token F1, so average improvement cannot hide harm to originally correct answers. Third, we find an asymmetric evidential boundary: adding the internal HGB score to the GbV-only policy passes a joint EM-improvement and Damage-reduction rule, whereas adding GbV to the HGB-only policy does not. The latter result is not evidence that GbV is useless, nor a formal comparison of the two incremental effects.

2 Related work

Adaptive and corrective RAG. Adaptive-RAG selects among no-, single-, and multi-step retrieval from question complexity [9]. Self-RAG trains the generator to retrieve and critique its outputs [2], while CRAG evaluates retrieval quality and invokes corrective actions [20]. Verify-and-Edit verifies reasoning chains, retrieves evidence, and edits low-consistency outputs [22]. D2R-RAG diagnoses failures and selects repair actions under resource constraints [7]. Doctor-RAG localizes and repairs known failed agentic trajectories while reusing valid prefixes [10]. These systems change retrieval or generation behavior. Our policy receives two answers that have already been acquired and only selects the final output across the complete trace population. Its 5% allocation limits actions; it is not a 5% compute budget.

Predicting benefit and post-generation arbitration. Self-Knowledge Guided Retrieval learns when external retrieval is helpful [17]. Dado et al. directly predict the benefit of RAG over a no-RAG answer using retrieval, answer, and semantic-consistency signals [4]. Their version 3 includes a paired NLI difference, the two-answer Bert-Gen-2A predictor, and predicted-gain-driven Selective RAG evaluation (Sections 4.3 and 7). TrustMargin selects between Direct and RAG answers using likelihood views without training [19]. Counterfactual supervision has also been used to route search from paired no-search and forced-search outcomes [11]. Our candidate domain is original-RAG versus repaired-RAG, and our attribution question fixes supervised controls and a global switch count. The changed domain and evaluation protocol do not make two-candidate selection itself new. These works are conceptual comparators; we do not report a numerical comparison with their complete systems or claim superiority to them.

Verification and retrieval harm. Generate but Verify introduces Answering with Faithfulness and evaluates faithfulness prediction for downstream use [6]. We adapt its entailment-based evidence score to both branches and use the difference as one policy input. Because our outcomes are normalized answer correctness, we call this a paired GbV adaptation rather than a direct faithfulness evaluation. Work on detrimental or distracting passages motivates explicit harm accounting [1, 13]. Pair-ID further shows that response to paired evidence interventions is conditional on the reader [5]. Accordingly, our claims are restricted to the fixed reader and realized candidate pipeline.

3 Study design

3.1 Population and shared candidate acquisition

We use 2,000 questions each from HotpotQA [21], 2WikiMultiHopQA [8], and MuSiQue [16]. Selection used a prospectively fixed deterministic hash order after recorded-ID exclusion; this is not a probability sample. Every question is crossed with BM25 [15], BGE dense retrieval [18],

Table 1: Frozen study design. The pools are dataset-specific.

Component	Frozen setting
Question groups / traces	6,000 / 18,000
Datasets	HotpotQA, 2WikiMultiHopQA, MuSiQue; 2,000 questions each
Retrievers	BM25, BGE dense, hybrid RRF; one sibling trace per question
Dataset-specific pool sizes	19,352; 11,746; 23,618 documents (54,716 total)
Reader	Qwen2.5-3B-Instruct, revision <code>aa8e7253...04d1</code>
Development supervision	4,500 questions / 13,500 traces; 3,600 fit + 900 calibration
Fresh eligibility / denominator	4,267 / 18,000 traces
Action allocation	Global $K = 900$ for each switching policy; Keep selects zero
Primary resampling	20,000 dataset-stratified question-cluster draws

and reciprocal-rank fusion (RRF) [3], giving 6,000 question groups and 18,000 traces. Statistical resampling keeps the three retriever siblings of each question together.

Each dataset has its own deduplicated corpus built from selected benchmark contexts without reading answers, supporting facts, or decompositions. The pools contain 19,352 HotpotQA, 11,746 2WikiMultiHopQA, and 23,618 MuSiQue documents. They are bounded benchmark-derived pools, not full Wikipedia.

For each trace, the pipeline retrieves the top five documents E_0 and generates a_0 . It then generates a repair query from the question and E_0 , without using a_0 , retrieves to depth 50 with the same retriever, and inserts the first document absent from E_0 . Original ranks 1–4 are retained and rank 5 is replaced to create E_1 , from which the reader generates a_1 . All policies receive the same (a_0, a_1) pair.

The reader is Qwen/Qwen2.5-3B-Instruct [14], pinned at revision `aa8e72537993ba99e69dfaafa59ed015b17504d1`. The dense encoder is BAAI/bge-base-en-v1.5 at revision `a5beb1e3e68b9ab74eb54cfd186867f64f240e1a`. Generation uses the frozen chat template, greedy decoding, seed 20260828, at most 48 answer tokens, at most 64 repair-query tokens, and a 16,000-character evidence budget. A prospectively selected 180-trace replay matches the accepted acquisition exactly.

3.2 Scores and policies

The shared label-free score matrix contains 11 numeric inputs. Seven are outputs of byte-fixed historical upstream estimators: state-symmetric HGB, state-symmetric logistic, no-cross-state, no- B , no-evidence-change, no-answer-form, and ordinary compact logistic. Three are fixed rules: B -rule, higher-own-likelihood, and likelihood margin. The eleventh is the paired GbV margin.

HGB abbreviates histogram-based gradient boosting and denotes a frozen historical pairwise selector, not a model fitted on the current 4,500-question development cohort. Let L_{ij} be the reader’s mean teacher-forced log probability for answer a_i under evidence E_j . The selector receives 48 state-symmetric features: eight repaired-minus-original branch differences and their interactions with five pair-context variables. Its historical binary target is one for an incorrect a_0 followed by a correct a_1 and zero for the reverse transition; pairs in which both answers have the same EM status do not enter that fit. Training augments every (x, y) with $(-x, 1 - y)$, and the saved HGB probabilities for x and $-x$ are symmetrized at scoring time. A larger HGB score therefore means stronger support for choosing a_1 . The complete feature dictionary, formula, fixed fit parameters, training scope, and provenance boundary appear in Appendix E, “Frozen historical HGB score.”

For the paired adaptation of GbV Post-Answering NLI, the verifier assigns the maximum

Shared candidate acquisition occurs before every policy decision

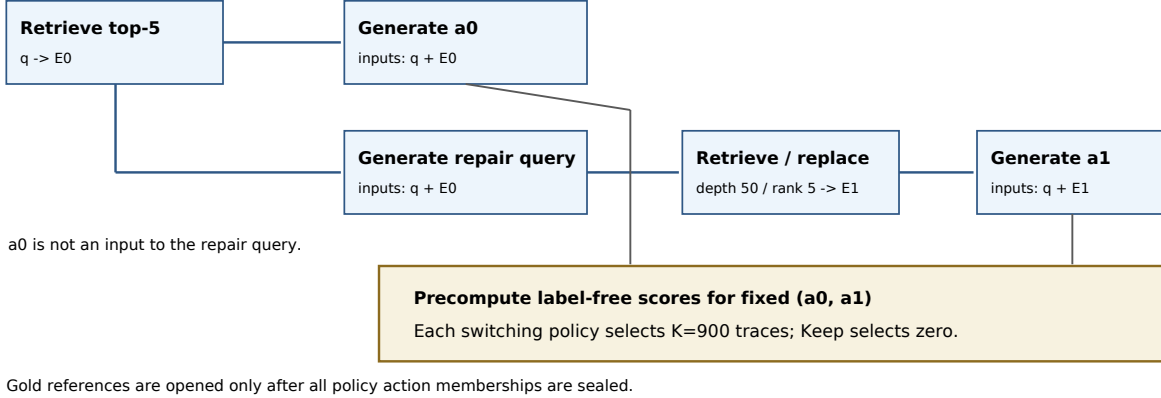


Figure 1: Shared paired candidate acquisition and downstream fixed-batch selection. Retrieval, repair-query generation, and both answers are acquired before policy allocation. Gold references are opened only after action memberships are sealed.

entailment probability over evidence chunks for each branch, $F_0(a_0, E_0)$ and $F_1(a_1, E_1)$; the policy input is $F_1 - F_0$. The verifier is MoritzLaurer/deberta-v3-large-zeroshot-v2.0 at revision 5a4338ab2151dc8db04ad53b42b6153382bf4f99, evaluated in FP32 with 512 tokens, 20-word chunk overlap, and entailment label index zero. The branch difference, revision/runtime, eligibility mask, and global allocation are project choices, not an author-supplied paired-policy formula or author-code reproduction. Base scoring also uses BGE answer-pair embeddings and four Qwen teacher-forced likelihood cells. These computations precede evaluation-outcome access.

The full panel contains Keep; raw HGB; raw paired GbV; ROA-FULL; ROA-NOGBV; HGB+GbV_R; HGB-only_R; GbV-only_R; and a sealed historical V2 policy. The subscript *R* denotes a fitted Recovery head. The five fitted policies are supervision matched:

- ROA-FULL uses all 11 numeric inputs, 11 missingness flags, and three retriever indicators (25 columns);
- ROA-NOGBV removes the GbV input and flag (23 columns);
- HGB+GbV_R uses HGB and the paired GbV margin, their missingness flags, and retriever indicators (7 columns);
- HGB-only_R and GbV-only_R each use one score, one missingness flag, and retriever indicators (5 columns).

Raw HGB, raw GbV, and V2 have different upstream training or rule dependencies and serve as context rather than matched attribution controls. Raw HGB ranks by the frozen historical score directly. HGB-only_R instead fits a current Recovery head using that one score, its missingness flag, and retriever indicators; HGB+GbV_R adds the paired GbV margin and its missingness flag. Both current heads are ordinary calibrated logistic regression, and neither refits the historical HGB estimator.

3.3 Matched supervision and allocation

All five heads use the same 4,500 development questions and 13,500 traces. Within each dataset, deterministic hash order assigns 1,200 questions to fitting and 300 disjoint questions to calibration;

retriever siblings stay together. Fit-only medians, means, and population standard deviations are used for preprocessing. Missingness and retriever indicators are unscaled.

The binary target is Recovery, defined by normalized EM changing from incorrect for a_0 to correct for a_1 . This target does not directly optimize Net or impose a Damage constraint; Damage reduction is an evaluated outcome. Each base head is L2 logistic regression with $C = 1$, `lbfgs`, no class weights, `max_iter`=5000, and tolerance 10^{-4} . A separate Platt logistic model with $C = 10^6$, `lbfgs`, `max_iter`=2000, and tolerance 10^{-4} calibrates disjoint calibration-set logits. The five base/calibration bundles account for ten of 185 total scientific fits in the complete project history; no model was refit after fresh outcomes were observed.

The fresh evaluation retains all 18,000 traces. A common eligibility mask contains 4,267 traces. The 13,732 traces with equal normalized answers and one trace with an empty original answer are forced to Keep for every policy. Every switching policy ranks eligible traces by its frozen score and receives the same global allocation $K = \text{round}(0.05 \times 18,000) = 900$. Ties use `(dataset, retriever, sample_id)`. There are no dataset quotas, retriever quotas, tuned thresholds, or post-outcome action changes.

3.4 Gold firewall, outcomes, and inference

For the evaluation cohort, candidate acquisition, all scores, the eligibility mask, and complete action memberships were sealed before reference access. A separate mapper then evaluated 6,000 selected reference sets (6,863 strings) against 36,000 fixed answers and emitted numeric EM/F1 outcomes only. An independent method-blind implementation recomputed all 72,000 scalars. This establishes the documented order within the authenticated execution graph, not the absence of unknown activity outside that graph.

For a policy action, Recovery is a_0 incorrect and a_1 correct; Damage is a_0 correct and a_1 incorrect; Neutral contains the other two transitions and need not mean unchanged token F1. Net equals Recovery minus Damage. EM and token F1 use the full 18,000-trace denominator. Damage rate is $100 \times \text{Damage} / 18,000$ percent, and the EM difference from Keep is $100 \times (\text{Recovery} - \text{Damage}) / 18,000$ percentage points.

The primary family contains three ordered comparisons and two endpoints per comparison: EM difference and Damage-rate difference. We use 20,000 dataset-stratified question-cluster bootstrap draws with NumPy `default_rng(20260926)`, keeping all retriever siblings together and reallocating the global top- K actions in each draw. Percentile endpoints use quantiles 0.05/12 and $1 - 0.05/12$ with linear interpolation, applying a two-sided Bonferroni adjustment to the six-endpoint family. The joint rule passes only when the adjusted EM range is strictly above zero and the adjusted Damage range is strictly below zero. A secondary sensitivity analysis reuses the draws while holding observed action memberships fixed; it does not enforce exactly 900 action copies within each draw. Intervals condition on the fitted heads and realized candidate pools. They exclude uncertainty from model fitting, calibration, candidate generation, and pool construction, and do not provide exact finite-sample coverage or guarantees for new readers or domains.

4 Results

4.1 Complete policy panel

Table 2 reports every policy. Keep has an EM of 17.9889% and token F1 of 22.7207%. Each switching policy executes 900 actions. Their EM gains over Keep range from 1.6278 to 1.9389

Table 2: Complete fixed-allocation results. EM, F1, and Damage rate are percentages; ΔEM and ΔF1 are differences from Keep in percentage points. Actions, Recovery, Damage count, Neutral, and Net are trace-event counts. All rates use 18,000 traces. Keep takes no action.

Policy	Act.	Rec.	Dmg.	Neu.	Net	EM (%)	ΔEM (pp)	F1 (%)	ΔF1 (pp)	Dmg. rate (%)
Keep	0	0	0	0	+0	17.9889	+0.0000	22.7207	+0.0000	0.0000
HGB	900	366	32	502	+334	19.8444	+1.8556	25.1593	+2.4386	0.1778
GbV	900	318	25	557	+293	19.6167	+1.6278	24.8928	+2.1722	0.1389
ROA-FULL	900	371	22	507	+349	19.9278	+1.9389	25.2988	+2.5781	0.1222
ROA-NOGBV	900	355	38	507	+317	19.7500	+1.7611	25.0321	+2.3115	0.2111
HGB+GbV _R	900	352	11	537	+341	19.8833	+1.8944	25.2780	+2.5574	0.0611
HGB-only _R	900	357	26	517	+331	19.8278	+1.8389	25.1323	+2.4117	0.1444
GbV-only _R	900	323	24	553	+299	19.6500	+1.6611	24.9344	+2.2137	0.1333
V2	900	361	28	511	+333	19.8389	+1.8500	25.1712	+2.4506	0.1556

Table 3: Primary ordered comparisons. Ranges are multiplicity-adjusted percentiles from 20,000 dataset-stratified question-cluster bootstrap draws with top- K actions reallocated in each draw. All values are percentage points.

Ordered comparison	ΔEM	Adjusted range	ΔDamage	Adjusted range	Joint rule
ROA-FULL – HGB+GbV _R	+0.0444	[-0.1333, +0.2037]	+0.0611	[+0.0000, +0.1278]	Not met
HGB+GbV _R – HGB-only _R	+0.0556	[-0.1556, +0.3167]	-0.0833	[-0.1833, -0.0056]	Not met
HGB+GbV _R – GbV-only _R	+0.2333	[+0.0722, +0.4333]	-0.0722	[-0.1444, -0.0278]	Met

percentage points, while their Damage rates range from 0.0611% to 0.2111%. All eight comparisons with Keep are descriptive and are outside the frozen primary interval family.

HGB+GbV_R records 352 Recoveries and 11 Damages (Net 341), yielding 19.8833% EM and 25.2780% token F1. Among the eight equal-action switching policies it has the lowest observed full-population Damage count and rate. Keep, which takes no actions, has zero Damage. ROA-FULL has the highest point EM and F1, but its 371 Recoveries are accompanied by 22 Damages; point ranking alone does not establish advancement.

4.2 Primary comparisons

Table 3 gives the complete primary family. Relative to GbV-only_R, HGB+GbV_R improves EM by 0.2333 percentage points and reduces Damage by 0.0722 points. The adjusted ranges, [0.0722, 0.4333] and [-0.1444, -0.0278], exclude zero in the required directions, so this comparison passes the joint rule. In event counts, the difference is 42 more correct final answers and 13 fewer Damage events across 18,000 traces.

Relative to HGB-only_R, HGB+GbV_R changes EM by 0.0556 points with an adjusted range of [-0.1556, 0.3167]. Its Damage difference is -0.0833 points with range [-0.1833, -0.0056]. Because the EM range crosses zero, the joint rule does not pass. This comparison cannot support a claim that adding GbV to HGB jointly improves accuracy and harm.

ROA-FULL minus HGB+GbV_R is 0.0444 points for EM [-0.1333, 0.2037] and +0.0611 points for Damage [0.0000, 0.1278]. It also fails the joint rule, and the Damage direction is not favorable. The 25-feature model therefore does not establish an advance over the simpler two-signal policy.

The primary comparisons test different increments. The positive comparison adds HGB to a GbV-only head; the inconclusive comparison adds GbV to an HGB-only head. One passing and one non-passing comparison is not a formal interaction test or evidence that the increments

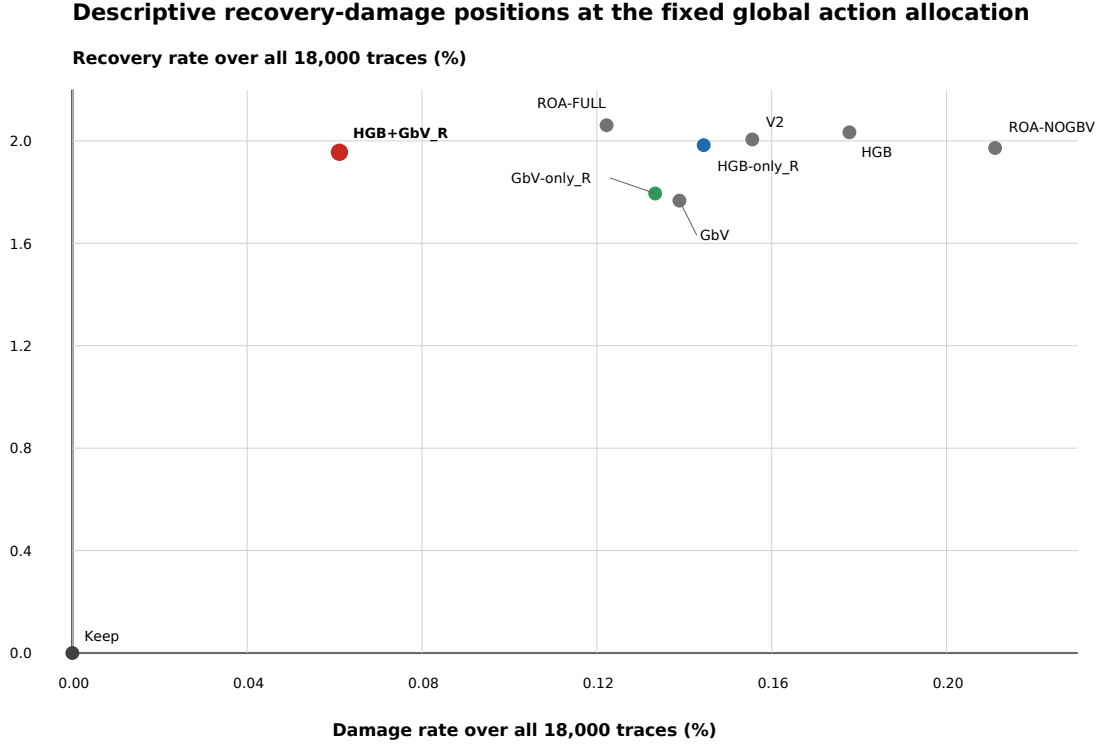


Figure 2: Descriptive full-population Recovery and Damage rates. The chart contains no uncertainty intervals; formal inference is restricted to Table 3.

differ causally. It also does not imply that GbV is unnecessary: the study was not designed as an equivalence or non-inferiority test.

4.3 Sensitivity and secondary breakdowns

Holding the observed action memberships fixed preserves the $\text{HGB} + \text{GbV}_R$ versus GbV-only_R joint decision: adjusted EM and Damage ranges are $[0.0389, 0.4278]$ and $[-0.1389, -0.0222]$. The HGB-only comparison remains non-passing: its EM range crosses zero and its Damage upper endpoint touches zero. The ROA-FULL comparison also remains non-passing. Dataset and retriever breakdowns are reported in Appendices B and C as descriptive consequences of the one global allocation. They were not separate primary families, and no favorable subgroup is promoted to a claim.

5 Workload and reproducibility evidence

All policies consume the same precomputed graph in this experiment. Candidate acquisition executes 18,000 original and 18,000 repair retrievals. Original retrieval comprises 12,000 BM25 and 12,000 dense components, 15,422 embedding forwards, and 54,716 document rows. Repair retrieval comprises 12,000 BGE query forwards without document re-embedding. The Qwen stage contains 54,000 generation receipts, 51,901,556 prompt tokens, and 826,362 generated tokens/output score steps. Base scoring records 2,250 answer-embedding and 16,932 likelihood forwards, with 136 likelihood cache hits; paired GbV records 8,534 NLI forwards over 42,946 premise/hypothesis pairs.

Preserved stage timers are 1,096.562 s for C2, 45,216.887 s for C3, 5,337.573 s for base scoring, 3,523.022 s for GbV scoring, and 157.658 s for fixed allocation. Their boundaries include different setup, caching, I/O, and audit work, so they are not summed or presented as end-to-end latency. Saved C4 allocator peaks are 8,153,970,176 bytes for base scoring and 2,956,706,304 bytes for GbV scoring; they occurred in different stages and are not additive.

No standalone policy deployment was built or timed. Canonical C2/C3 allocator peaks, per-trace latency distributions, one end-to-end timer, energy, and FLOPs were not measured. Consequently, the global 5% action count is not a claim of 5% retrieval, generation, latency, memory, or compute.

The numerical analysis is backed by hash-pinned aggregate files and independent implementations. A deterministic anonymous aggregate package contains the complete point estimates, intervals, claim summary, code/data availability statement, and verification scripts. Its historical candidate archive hash is `b785890366995c2943007f5635e622814bc7961dc7b3ded4adee0707dfb7ca8d`. That immutable candidate remains withheld because its notice predates the corrected third-party terms review. A corrected Apache-2.0-aware candidate has been rebuilt and independently validated; its release still requires institutional review, final authorization, and persistent archiving.

6 Threats to validity

Reader and population scope. The accepted result uses one 3B-parameter Qwen reader, three multi-hop QA datasets, three fixed retrieval conditions, and bounded benchmark-derived pools. This scale does not invalidate the within-reader matched comparison, but it prevents claims about reader size, other model families, unseen domains, full-corpus retrieval, or production use. A Phi extension produced authentic runtime artifacts but failed its prospectively fixed semantic validation gate; it supplies no effect result. A planned Mistral extension stopped before engineering and model execution.

Candidate and estimand scope. The repair operation replaces only rank 5 after a generated repair query. Other repair operators can change candidate quality and policy ranking. The estimand conditions on both candidates already existing, the fixed eligibility mask, and a global 900-action allocation. It does not measure an online threshold policy or compute-aware router.

Metric scope. Normalized EM is strict and can miss semantically correct variants; token F1 is included descriptively. Recovery and Damage measure correctness transitions rather than evidence faithfulness. The GbV component remains a paired adaptation of a faithfulness verifier. Damage uses the full population denominator, so its numerical rate should be interpreted with the event count and action count.

Inference scope. Question-cluster resampling respects retriever siblings and stratifies datasets, but the intervals condition on fixed fitted heads and realized pools. The study does not independently replicate the candidate-generation population. Secondary dataset/retriever cells are small, correlated views of the same global action allocation.

Provenance and contamination. The selected cohort has no dataset/ID overlap with a known 18,600-question exclusion union, but ID separation is not semantic decontamination and cannot rule out model pretraining exposure or unknown unlogged historical use. Benchmark-supplied contexts may also share dataset-specific structure with answers despite the field-restricted pool construction. Seven historical upstream estimators are byte-fixed and their saved-parameter computation is

replayed. A locally preserved pre-ROA-named package contains a byte-identical copy of the complete historical training manifest, but no external record independently certifies its timestamp; the original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain missing. Reproduction therefore authenticates the saved computation within checked environments, not a literal replay of every original training event. Appendix G records the distinct historical and current development populations.

7 Discussion

The main result is comparison dependent. Starting from a GbV-only fitted selector, adding the internal HGB score is supported by both endpoints. Starting from HGB-only, the available data do not establish that adding GbV jointly improves EM and Damage. This pattern can arise from redundancy, score quality, finite event counts, calibration, or the realized candidate set. The experiment does not distinguish these mechanisms.

The result also illustrates why unmatched comparisons can mislead. Raw scores and historical V2 do not share the same development recipe as the five fitted controls. Comparing the two-signal head only with raw GbV would mix feature value with supervision and calibration. Likewise, selecting ROA-FULL from the highest point EM would ignore uncertainty and its less favorable observed Damage. Matched single-signal heads isolate a more precise question even when the answer is negative.

For deployment, the study supports neither a universal fusion rule nor a cost saving. All candidates and scores were precomputed for fairness. A real system would need a prospectively timed standalone graph, an online allocation or threshold rule, drift monitoring, and validation on its own reader and corpus. Those engineering decisions are outside the present evidence.

8 Conclusion

On a fixed 6,000-question Qwen2.5-3B-Instruct cohort with three datasets, three retrievers, shared original/repaid candidates, matched supervision, and a global 5% action allocation, the HGB+GbV_R selector improves EM and reduces Damage relative to GbV-only_R. The same joint rule does not pass against HGB-only_R, and a larger feature stack does not advance over the two-signal selector. The contribution is the controlled attribution and its negative boundary, together with full-population harm accounting. It is not a new selector architecture or evidence of transfer beyond the evaluated reader and candidate pipeline.

Data and code availability

An anonymous, aggregate-only verification package has been built and independently validated. It contains no questions, answers, Gold strings, per-example outcomes, model weights, local paths, or identities. Apache License 2.0 is present for project-authored code, and a corrected license-aware aggregate candidate has been built and validated. The project code is held in a public GitHub repository, but the final submission revision and permanent archive identifier have not yet been frozen. Public distribution of the aggregate candidate remains withheld pending confirmation of the legal holder/year, institutional and third-party release review, final archive authorization, and a persistent identifier. No archive with a DOI or other persistent identifier has been released. The final submission will provide a frozen code link and immutable archive identifier after the owner and institutional release gates pass. Benchmark- or model-restricted evidence will be described

with its access conditions and supplied if requested through a channel designated or accepted by the handling editor, subject to upstream terms. The manuscript claims aggregate verification, not end-to-end public neural reproduction.

Declarations

Author identities, affiliations, contributions, funding, competing interests, institutional ethics wording, acknowledgements, and any required disclosure of writing assistance are maintained in the separate title-page/declarations template and must be completed by the responsible authors before submission.

References

- [1] Chen Amiraz, Florin Cuconasu, Simone Filice, and Zohar Karnin. The distracting effect: Understanding irrelevant passages in rag. In *Proceedings of ACL*, pages 18228–18258, 2025. doi: 10.18653/v1/2025.acl-long.892.
- [2] Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-rag: Learning to retrieve, generate, and critique through self-reflection. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=hSyW5go0v8>.
- [3] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of SIGIR*, pages 758–759, 2009. doi: 10.1145/1571941.1572114.
- [4] Or Dado, David Carmel, and Oren Kurland. Predicting the benefit of retrieval augmentation in open-domain question answering. *arXiv preprint arXiv:2604.07985v3*, 2026. doi: 10.1145/3799682.3840689. URL <https://arxiv.org/abs/2604.07985v3>.
- [5] Wenzhang Du. What would fix this rag failure? auditing counterfactual response with paired evidence interventions. *arXiv preprint arXiv:2608.08944*, 2026. URL <https://arxiv.org/abs/2608.08944>.
- [6] Simone Filice, Elad Haramaty, Guy Horowitz, Zohar Karnin, Liane Lewin-Eytan, and Alex Shtoff. Generate but verify: Answering with faithfulness in rag-based question answering. In *Proceedings of IJCNLP-AACL*, pages 1017–1037, 2025. doi: 10.18653/v1/2025.ijcnlp-long.56. URL <https://aclanthology.org/2025.ijcnlp-long.56/>.
- [7] Soroush Hashemifar, Havva Alizadeh Noughabi, Fattane Zarrinkalam, and Ali Dehghantanha. Diagnosing and repairing factual errors in rag under budget constraints. *arXiv preprint arXiv:2606.29377*, 2026. URL <https://arxiv.org/abs/2606.29377>.
- [8] Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of COLING*, pages 6609–6625, 2020. doi: 10.18653/v1/2020.coling-main.580.
- [9] Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong Park. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of NAACL-HLT*, pages 7036–7050, 2024. doi: 10.18653/v1/2024.naacl-long.389.

- [10] Shuguang Jiao, Chengkai Huang, Shuhan Qi, Xuan Wang, Yifan Li, Quanchi Weng, Lingchuan Liu, Xunliang Cai, and Lina Yao. Doctor-RAG: A failure-aware repair framework for agentic retrieval-augmented generation. *arXiv preprint arXiv:2604.00865v2*, 2026. URL <https://arxiv.org/abs/2604.00865v2>.
- [11] Minh Kim. When should llms search? counterfactual supervision for search routing. *arXiv preprint arXiv:2607.05752*, 2026. URL <https://arxiv.org/abs/2607.05752>.
- [12] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.
- [13] Philhoon Oh and James Thorne. Detrimental contexts in open-domain question answering. In *Findings of EMNLP*, pages 11589–11605, 2023. doi: 10.18653/v1/2023.findings-emnlp.776.
- [14] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024. URL <https://arxiv.org/abs/2412.15115>.
- [15] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/15000000019.
- [16] Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. Musique: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022. doi: 10.1162/tacl.a_00475.
- [17] Yile Wang, Peng Li, Maosong Sun, and Yang Liu. Self-knowledge guided retrieval augmentation for large language models. In *Findings of EMNLP*, pages 10303–10315, 2023. doi: 10.18653/v1/2023.findings-emnlp.691.
- [18] Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-Pack: Packed resources for general chinese embeddings. *arXiv preprint arXiv:2309.07597v5*, 2024. URL <https://arxiv.org/abs/2309.07597v5>.
- [19] Jingyan Xu, Hong Shi, Yi Shan, Penghui Liu, Yunhao Bai, Ningyuan Li, and Xueyang Liu. Trustmargin: Training-free arbitration between parametric memory and retrieved evidence in large language models. *arXiv preprint arXiv:2606.08397*, 2026. URL <https://arxiv.org/abs/2606.08397>.
- [20] Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*, 2024. URL <https://arxiv.org/abs/2401.15884>.
- [21] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. Hotpotqa: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of EMNLP*, pages 2369–2380, 2018. doi: 10.18653/v1/D18-1259.
- [22] Ruochen Zhao, Xingxuan Li, Shafiq Joty, Chengwei Qin, and Lidong Bing. Verify-and-edit: A knowledge-enhanced chain-of-thought framework. In *Proceedings of ACL*, pages 5823–5840, 2023. doi: 10.18653/v1/2023.acl-long.320.

Appendices

Appendix A. Fixed-action sensitivity

The primary analysis reallocates the global top- K actions within every question-cluster bootstrap draw. The following pre-specified sensitivity holds the observed action memberships fixed while reusing the same 20,000 draws and six-endpoint multiplicity adjustment.

Table 4: Fixed-action sensitivity. Values are percentage points.

Ordered comparison	Δ EM	Adjusted range	Δ Damage	Adjusted range	Joint rule
ROA-FULL – HGB+GbV _R	+0.0444	[-0.1333, +0.2278]	+0.0611	[+0.0000, +0.1278]	Not met
HGB+GbV _R – HGB-only _R	+0.0556	[-0.2000, +0.3111]	-0.0833	[-0.1722, +0.0000]	Not met
HGB+GbV _R – GbV-only _R	+0.2333	[+0.0389, +0.4278]	-0.0722	[-0.1389, -0.0222]	Met

Appendix B. Dataset breakdown

The cells below describe the one global allocation after partitioning by dataset. Each cell contains 6,000 traces. Actions, Recovery, Damage, and Net are counts; the EM/F1 differences from that cell’s Keep baseline are percentage points with the cell denominator. The cells do not reallocate 900 actions within each dataset and support no subgroup inference.

Cell	Policy	Act.	Rec.	Dmg.	Net	Δ EM (pp)	Δ F1 (pp)
2wikimultihopqa	Keep	0	0	0	+0	+0.0000	+0.0000
2wikimultihopqa	HGB	292	136	12	+124	+2.0667	+2.6822
2wikimultihopqa	GbV	319	131	11	+120	+2.0000	+2.6189
2wikimultihopqa	ROA-FULL	298	134	8	+126	+2.1000	+2.8632
2wikimultihopqa	ROA-NOGBV	300	127	15	+112	+1.8667	+2.5163
2wikimultihopqa	HGB+GbV _R	297	129	4	+125	+2.0833	+2.8465
2wikimultihopqa	HGB-only _R	290	133	9	+124	+2.0667	+2.6644
2wikimultihopqa	GbV-only _R	318	132	10	+122	+2.0333	+2.6489
2wikimultihopqa	V2	313	139	10	+129	+2.1500	+2.8482
hotpotqa	Keep	0	0	0	+0	+0.0000	+0.0000
hotpotqa	HGB	323	132	16	+116	+1.9333	+2.6733
hotpotqa	GbV	303	98	10	+88	+1.4667	+2.1477
hotpotqa	ROA-FULL	315	122	10	+112	+1.8667	+2.6628
hotpotqa	ROA-NOGBV	316	121	17	+104	+1.7333	+2.4339
hotpotqa	HGB+GbV _R	309	121	7	+114	+1.9000	+2.7132
hotpotqa	HGB-only _R	286	119	12	+107	+1.7833	+2.4747
hotpotqa	GbV-only _R	300	99	10	+89	+1.4833	+2.1814
hotpotqa	V2	317	123	14	+109	+1.8167	+2.5511
musique	Keep	0	0	0	+0	+0.0000	+0.0000
musique	HGB	285	98	4	+94	+1.5667	+1.9603
musique	GbV	278	89	4	+85	+1.4167	+1.7499

Cell	Policy	Act.	Rec.	Dmg.	Net	Δ EM (pp)	Δ F1 (pp)
musique	ROA-FULL	287	115	4	+111	+1.8500	+2.2084
musique	ROA-NOGBV	284	107	6	+101	+1.6833	+1.9843
musique	HGB+GbV _R	294	102	0	+102	+1.7000	+2.1124
musique	HGB-only _R	324	105	5	+100	+1.6667	+2.0959
musique	GbV-only _R	282	92	4	+88	+1.4667	+1.8109
musique	V2	270	99	4	+95	+1.5833	+1.9524

Appendix C. Retriever breakdown

The cells below describe the same global allocation after partitioning by retriever. Each cell contains 6,000 traces. Counts and percentage-point differences have the same meaning as above, using the retriever-cell denominator. These cells support no retriever-specific superiority claim.

Cell	Policy	Act.	Rec.	Dmg.	Net	Δ EM (pp)	Δ F1 (pp)
bm25	Keep	0	0	0	+0	+0.0000	+0.0000
bm25	HGB	386	149	15	+134	+2.2333	+2.9645
bm25	GbV	318	111	7	+104	+1.7333	+2.3527
bm25	ROA-FULL	291	119	4	+115	+1.9167	+2.5157
bm25	ROA-NOGBV	296	117	8	+109	+1.8167	+2.2911
bm25	HGB+GbV _R	300	118	2	+116	+1.9333	+2.5663
bm25	HGB-only _R	212	100	2	+98	+1.6333	+2.0095
bm25	GbV-only _R	295	109	5	+104	+1.7333	+2.3324
bm25	V2	391	154	10	+144	+2.4000	+3.0984
dense	Keep	0	0	0	+0	+0.0000	+0.0000
dense	HGB	282	130	8	+122	+2.0333	+2.5925
dense	GbV	276	101	6	+95	+1.5833	+2.0890
dense	ROA-FULL	305	129	5	+124	+2.0667	+2.7886
dense	ROA-NOGBV	307	122	15	+107	+1.7833	+2.4558
dense	HGB+GbV _R	312	125	3	+122	+2.0333	+2.7335
dense	HGB-only _R	395	150	13	+137	+2.2833	+3.0715
dense	GbV-only _R	305	112	7	+105	+1.7500	+2.3119
dense	V2	271	113	9	+104	+1.7333	+2.3713
hybrid	Keep	0	0	0	+0	+0.0000	+0.0000
hybrid	HGB	232	87	9	+78	+1.3000	+1.7588
hybrid	GbV	306	106	12	+94	+1.5667	+2.0747
hybrid	ROA-FULL	304	123	13	+110	+1.8333	+2.4302
hybrid	ROA-NOGBV	297	116	15	+101	+1.6833	+2.1875
hybrid	HGB+GbV _R	288	109	6	+103	+1.7167	+2.3724
hybrid	HGB-only _R	293	107	11	+96	+1.6000	+2.1540
hybrid	GbV-only _R	300	102	12	+90	+1.5000	+1.9970
hybrid	V2	238	94	9	+85	+1.4167	+1.8821

Appendix D. Policy construction details

All five supervised heads use 3,600 fit questions and 900 disjoint calibration questions, with three retriever siblings retained within a question group. Numeric imputation and standardization use fit data only. Missing-value indicators and one-hot retriever indicators are unscaled. The base classifiers and Platt calibrators use fixed logistic-regression settings and are never refit after fresh outcomes are opened. Raw HGB, raw GbV, and V2 are included for descriptive context but are not supervision-matched attribution controls.

Appendix E. Frozen historical HGB score

Identity and target. HGB means the historical *histogram-based gradient-boosting* pairwise selector implemented with `HistGradientBoostingClassifier`. It is a frozen upstream estimator and is not the HGB-only_R head fitted in the current experiment. Its historical development universe contains 7,200 labeled traces from 4,800 question groups and 1,539 available pair-feature records. Only 601 informative traces from 518 question groups enter the final fit. For normalized exact-match outcomes $c_i \in \{0, 1\}$ of answer a_i , the historical target is

$$y = \begin{cases} 1, & (c_0, c_1) = (0, 1), \\ 0, & (c_0, c_1) = (1, 0). \end{cases}$$

Pairs with $(c_0, c_1) = (0, 0)$ or $(1, 1)$ have no preference label and are excluded from HGB fitting. Thus HGB learns which member of an already acquired answer pair is correct when exactly one member is correct; it does not learn the current experiment’s full-population Recovery target from all eligible traces.

Likelihood cells and branch features. For $i, j \in \{0, 1\}$, L_{ij} is the mean teacher-forced token log probability of the fixed answer a_i when the reader is conditioned on the question and evidence set E_j . Hence L_{00} and L_{11} are the two own-evidence likelihoods, while L_{01} and L_{10} score each answer under the other branch’s evidence. For branch i , define the eight-component vector

$$\phi_i = (\ell_i^{\text{own}}, \ell_i^{\text{cross}}, \ell_i^{\text{own}} - \ell_i^{\text{cross}}, n_i, u_i^+, u_i^-, s_i^+, s_i^-),$$

where $\ell_0^{\text{own}} = L_{00}$, $\ell_0^{\text{cross}} = L_{01}$, $\ell_1^{\text{own}} = L_{11}$, and $\ell_1^{\text{cross}} = L_{10}$. Here n_i is answer token length; u_i^+ and u_i^- are multiset token F1 between a_i and the added and removed document text; and s_i^+ and s_i^- are `SequenceMatcher` ratios between normalized a_i and those texts. The last two historical field names contain “semantic compatibility,” but their frozen implementation is normalized string-sequence similarity rather than a neural semantic model.

Let $\delta = \phi_1 - \phi_0$. The five pair-context variables are

$$z = (o_E, d_S, n_E, t_A, s_A),$$

where o_E is the Jaccard overlap of document IDs in E_0 and E_1 ; d_S is added-document retrieval score minus removed-document retrieval score; n_E is one minus the largest token-set Jaccard similarity between the added document and any document in E_0 ; t_A is token F1 between a_0 and a_1 ; and s_A is their saved BGE answer-pair similarity. The HGB input is

$$x = [\delta; \delta \otimes z] \in \mathbb{R}^{8+8 \times 5} = \mathbb{R}^{48}.$$

This compact construction expands to eight branch differences and forty difference–context interactions. Dataset and retriever identities are absent from the historical HGB input.

State symmetry, model, and score direction. The fit duplicates each historical example as (x, y) and $(-x, 1 - y)$. The fixed classifier uses `max_iter=300`, `max_leaf_nodes=15`, learning rate 0.08, L2 regularization 1.0, and final-fit seed 20261834; no feature scaling or hyperparameter search is applied to this final HGB fit. If $p_H(1 | x)$ is its class-one probability, the stored repair-preference score is

$$h(x) = \frac{1}{2}\{p_H(1 | x) + 1 - p_H(1 | -x)\}.$$

Consequently $h(-x) = 1 - h(x)$ up to numerical precision, and a larger $h(x)$ favors replacing a_0 with a_1 . Raw HGB ranks eligible traces directly by $h(x)$. HGB-only_R is a separate current L2-logistic Recovery head whose inputs are $h(x)$, one missingness flag, and three retriever indicators; HGB+GbV_R adds the paired GbV margin and its flag. The two current heads share the current fit/calibration recipe and never refit HGB.

Source and provenance boundary. The authenticated historical definitions are `src/mars/state_symmetric.py` (SHA-256 3724b5ac77722b70cabdf2589379d5584942f34ec81f3f5a04f88e0665f8f717) and the model binding in `src/mars/full_experiment.py` (SHA-256 65713509ef6c8d094bb44ad2c47b0238ceade607616d3e729b31c56096326d50). The original saved HGB artifact is 639,652 bytes with SHA-256 9245170f855435b5603b04013bd4fbab79a75d2761853a012ab9f3259600bf6e. A source-assembled replay reproduced all 1,539 HGB scores with zero numerical error. Its serialized file was not byte-identical because the saved bin mapper recorded one CPU-thread metadata value instead of six; all observed learned arrays and tree fields matched. The original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable, so this evidence authenticates the saved computation and reconstructed fit, not the literal historical execution event.

Appendix F. Answer metrics

Normalization lowercases text, removes ASCII punctuation and the English articles **a**, **an**, and **the**, and collapses whitespace. EM is one if the normalized prediction equals any allowed reference. Token F1 uses multiset token overlap and takes the maximum over allowed references. Unequal answers involving **yes**, **no**, or **noanswer** receive zero F1; an empty token sequence scores one only against another empty sequence. The method-blind independent metric implementation applies these same conventions. No new reference evaluation is performed for manuscript preparation.

Appendix G. Data and historical-development scope

The evaluation source frames are HotpotQA validation, 2WikiMultiHopQA development, and MuSiQue training. The last label describes the public benchmark split, not membership in the current selector’s fit set. A prospectively fixed hash order selected 2,000 questions per dataset after excluding the recorded 18,600-question union; the result is a deterministic frame-scoped cohort, not a probability sample or semantic-decontamination certificate.

The seven historical upstream estimators have reconstructed membership of 601 fit traces / 518 question groups within a 7,200-trace / 4,800-question historical development universe. Those 4,800 groups and the 3,000 earlier V2 questions belong to the exclusion union and have zero recorded dataset/ID overlap with the current five-head development cohort. Only the five current heads share the 4,500-question, 3,600-fit / 900-calibration recipe. Upstream learning histories are part of the fixed signal dependency, so matched current heads do not imply equal total historical label budgets for every signal.

Appendix H. Provenance and stopped extensions

Saved estimator bytes, reconstructed historical membership, numerical replay, pre-label scoring witnesses, action memberships, outcome mapping, aggregate analysis, and independent validators are preserved under hashes. A locally preserved pre-ROA-named package contains a byte-identical copy of the complete historical training manifest, but its timestamp has no independent external certification. Seven original per-estimator fit-time ID/matrix receipts and an independent original-fit witness remain unavailable. These disclosures do not exclude unknown unlogged activity. The Qwen/BGE replay covers a prospectively selected 180 of 18,000 traces; downstream saved neural witnesses are fully checked, but this is not a second complete neural-forward run.

A Phi reader extension acquired artifacts but failed its fixed semantic validation because 6,573 of 9,120 BF16 query vectors exceeded the 10^{-3} tolerance, with maximum deviation 0.0041683525. It was excluded before policy fitting and fresh effect analysis. A Mistral extension stopped at protocol review before engineering or model execution. Neither is a failed reader-effect experiment, and neither supports a robustness claim.

Appendix I. Workload disclosure

Stage timers cover distinct receipt boundaries and must not be summed: C2 1,096.562 s; C3 45,216.887 s; base scoring 5,337.573 s; GbV scoring 3,523.022 s; allocation 157.658 s. Separate saved C4 peaks of 8,153,970,176 and 2,956,706,304 bytes are also non-additive. The bounded replay separately contains 540 generation receipts, 518,966 prompt tokens, and 8,441 generated tokens/forwards. It is a reproduction witness, not an independent workload estimate. Standalone policy and per-generation latency, a uniform end-to-end timer, canonical C2/C3 peaks, a simultaneous whole-system peak, C2/C3 BGE token totals, generation input-token touches per autoregressive forward, FLOPs, and energy were not measured. Missing values are not zero; prompt or output-token totals are not FLOP estimates.

Appendix J. Reproduction boundary

The anonymous aggregate candidate permits deterministic checking of published numbers, interval signs, joint decisions, and file hashes. It excludes raw benchmark text, model weights, local paths, per-example outcomes, and private immutable ledgers. This supports aggregate reporting verification. It does not yet support public end-to-end regeneration of retrieval, answers, neural scores, or upstream training.